(iv) Replace h by x, and k by y, in the resulting equation. The resulting equation is the equation of the locus of point P.

Example 6.1 Find the locus of a point which moves such that its distance from the x-axis is equal to the distance from the y-axis.

Solution:

Let $P\,(h,k)$ be a point on the locus.
Let A and B be the foot of the perpendiculars drawn from the point P on the x-axis and the y-axis respectively.
Therefore P is $(OA, OB) = (BP, AP) = (h,k)$
Given that $AP = BP$

$$\Rightarrow k = h$$

replacing h and k by substituting $h = x$ and $k = y$
The locus of P is, $y = x$, is a line passing through the origin

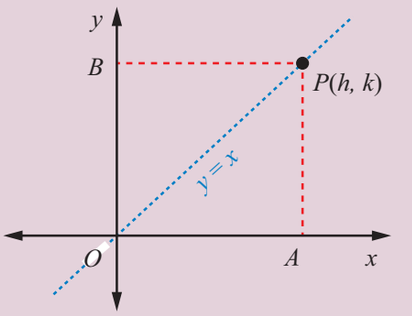


Figure 6.9

Example 6.2 Find the path traced out by the point $\left(ct, \frac{c}{t}\right)$, here $t \neq 0$ is the parameter and c is a constant

Solution:

Let $P\,(h,k)$be a point on the locus. From the given information, we have $h = ct$ and $k = \frac{c}{t}$.
To eliminate t, taking product of these two equations

$$(h)\,(k) = (ct)\left(\frac{c}{t}\right) \quad \Rightarrow \quad hk = c^2$$

Therefore, the required locus is $xy = c^2$

Example 6.3 Find the locus of a point P moves such that its distances from two fixed points $A\,(1,0)$ and $B\,(5,0)$, are always equal.

Solution:

Given that $A\,(1,0)$ and $B\,(5,0)$
Let $P\,(h,k)$ be any point on the required path.
From the information we have $AP = BP$
That is

$$\sqrt{(h-1)^2 + (k-0)^2} \;=\; \sqrt{(h-5)^2 + (k-0)^2} \Rightarrow h \;=\; 3$$

Therefore the locus of P is $x = 3$, which is a straight line parallel to the y-axis.

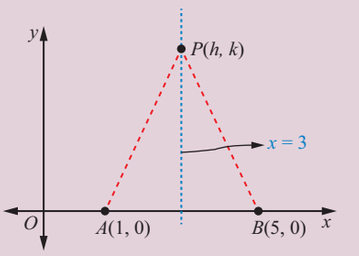


Figure 6.10